Measuring the Specific Heat Capacity of Aluminium

Lab report by Lorcan Nicholls, 16th September 2020

Abstract

The *specific heat capacity (SHC)* is a bulk quantity of a material defining the energy required to raise 1 kg of the substance by 1 °C. In this report, a method for determining the SHC of aluminium is presented with the result and uncertainty analysed. The value obtained was 881 ± 55 J kg^{-1} K^{-1}. According to an online source[1], the true value is 910 J kg^{-1} K^{-1}, which lies within our range.

Introduction

Thermal energy is the kinetic energy of the vibrating particles in a material. Thermal energy transfers are governed by the laws of thermodynamics: the first law of which, states that the sum of the thermal energy changes in a closed system is zero. This is a direct result of conservation of energy, applied to thermal energy.

Theory

When E joules of thermal energy is transferred to a material of mass m kg and specific heat capacity c J kg^{-1} K^{-1}, the increase in its temperature ΔT is related by

$E = mc\Delta T$ (Equation 1).

When the energy is provided by an electric heater, the energy transferred at a voltage V and current I for time t is given by

$E = IVt$ (Equation 2)

Equating (1) and (2),

$IVt = mc\Delta T$

If ΔT is taken as the dependent variable while t is varied, and all other quantities are controlled (i.e. kept constant), then the equation can be rewritten,

$c = (IV/m)(t/\Delta T)$ (Equation 3)

(3) shows c as being proportional to the gradient of a graph of time against temperature increase, which allows the experimental determination of c by continuous heating and measuring at specified time intervals.

Heat loss to the surroundings is minimised by insulating the material and covering the heating element in highly heat-conductive oil.

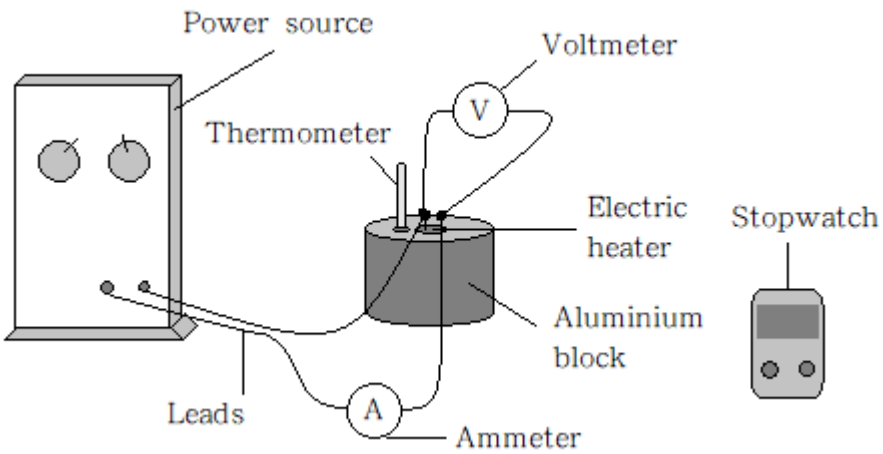
Setup and Method

The apparatus we used was:

- 1 kg cylindrical block of aluminium metal
- Voltmeter and ammeter
- Electric heater, insulated
- Electrical wires
- Power supply
- Thermometer
- Timer

The setup used is shown in *Figure 1*.

Figure 1



Once set up, the procedure we followed was:

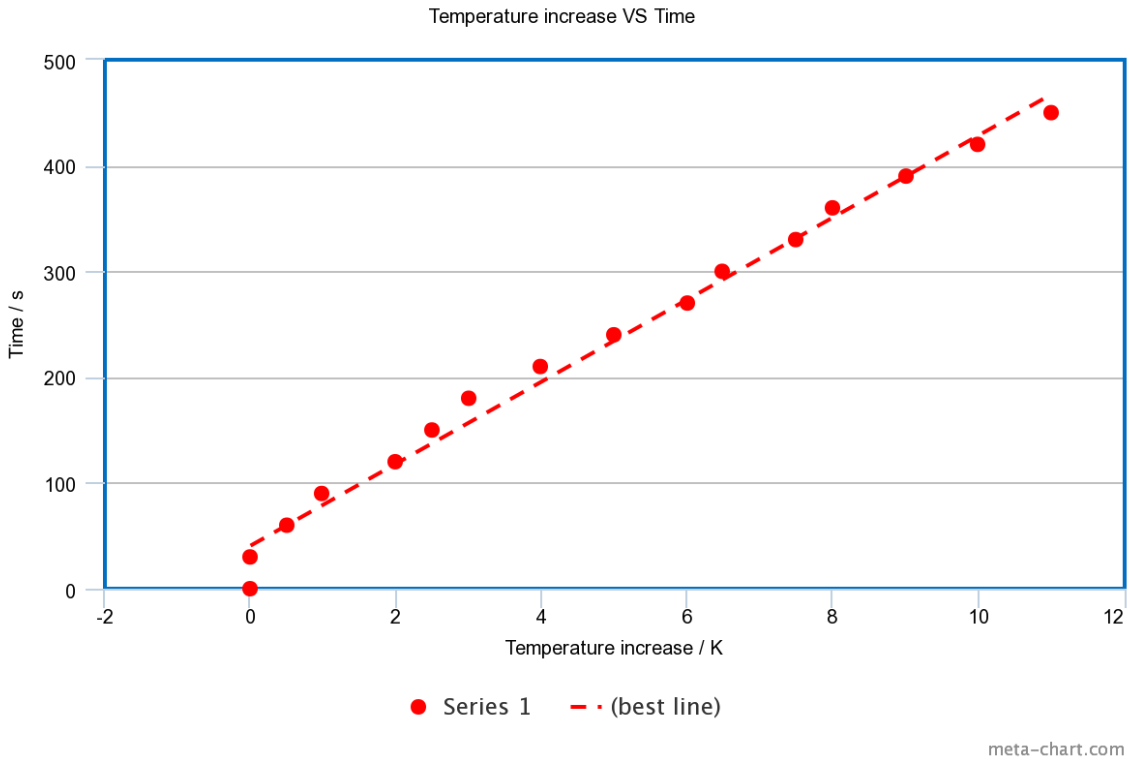
1. Measure the initial temperature of the aluminium
2. Turn the power supply on, with the voltage set to the maximum
3. Start the stopwatch
4. Record the mean value of the voltmeter and ammeter
5. When the stopwatch reaches 30 seconds, measure the temperature on the thermometer.
6. Repeat taking measurements from the thermometer every 30 seconds until 15 minutes have passed.
7. Calculate the power of the supply.
8. Plot a graph of results, time against temperature increase, and draw a line of best fit
9. Calculate the gradient and hence obtain a value for the SHC of aluminium.

Results

The temperature difference between the initial and at measurement was calculated at regular 30 second intervals for 7.5 minutes. The results are shown in *Table 1*.

Table 1

Time / s	0	30	60	90	120	150	180	210	240
Temperature increase / C	0	0	0.5	1.0	2.0	2.5	3.0	4.0	5.0
Time / s	270	300	330	360	390	420	450		
Temperature increase / C	6.0	6.5	7.5	8.0	9.0	10.0	11.0		



The best fit line extends from times of 40.67 to 467.54 over a temperature range of 11. This gives a gradient of 38.8. Since V = 8.44, I = 2.69 and m = 1.00, the proportionality factor in Equation 3 is VI/m = 22.7. Therefore, our value of c, is 38.8 * 22.7 = 881 J kg K^{-1}.

The percentage uncertainty in c is the sum of the percentage uncertainties in the variables. Our absolute uncertainties in V and I were 8.44 ± 0.08 and 2.69 ± 0.02 respectively, giving percentage uncertainty due to random variation as 1.7%. The resolution of the thermometer was ± 0.5, giving an average uncertainty of 4.5%. Therefore, the total percentage was 6.2%. This leads to our value of c lying in the interval 881 ± 55 J kg K^{-1}.

Conclusions

We obtained a range for *c* of 881 ± 55 J kg K^{-1}, a bound within which the reported true value of 910 lies within. Therefore, we claim our experiment is accurate and valid. The accuracy of this experiment could be improved by automating the process of recording the temperature, using a digital thermometer and having a computer record the data instead of a human experimenter. With the given apparatus, we believe our uncertainty and method is justified and produced a reliable result.

References

[1] The Engineering Toolbox: *Heat capacity of metals* (Accessed 16th September 2019)

https://www.engineeringtoolbox.com/specific-heat-metals-d_152.html